

CHAPTER 2: INTELLIGENT AGENTS

INTRODUCTION TO ARTIFICIAL INTELLIGENT (AI)



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Intelligent Agents

INTRODUCTION TO ARTIFICIAL INTELLIGENCE



Outline

- Agents and Environment
- **PEAS** (**P**erformance Measure, **E**nvironment, **A**ctuators, **S**ensors)
- The Nature of the Environment
- The Structure of Agent Types

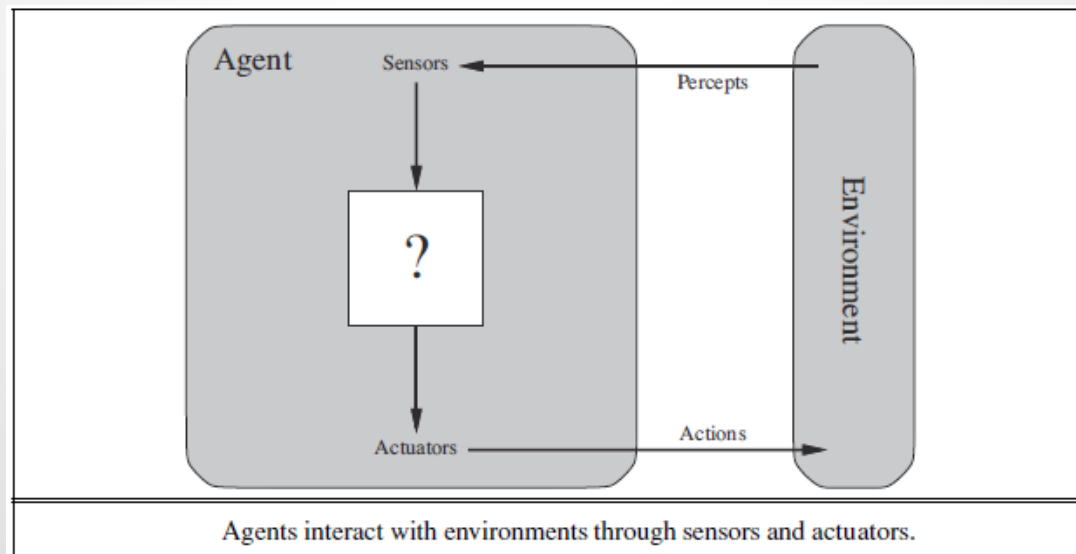
Agents and Environment

INTRODUCTION TO ARTIFICIAL INTELLIGENCE



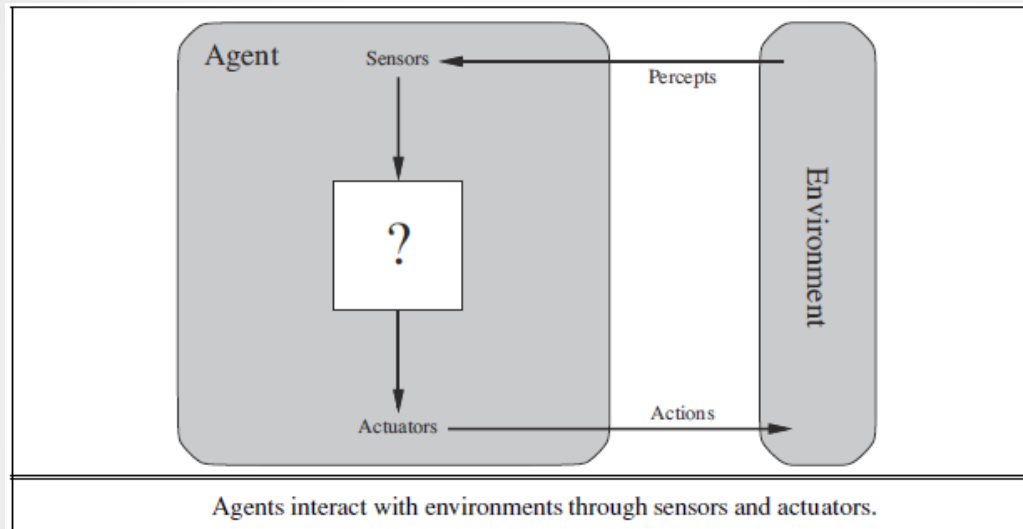
An **agent** is a thing that acts (possibly rationally).

An agent perceives its **environment** through **sensors** and acts upon the **environment** through **actuators**.



Agents and Environment

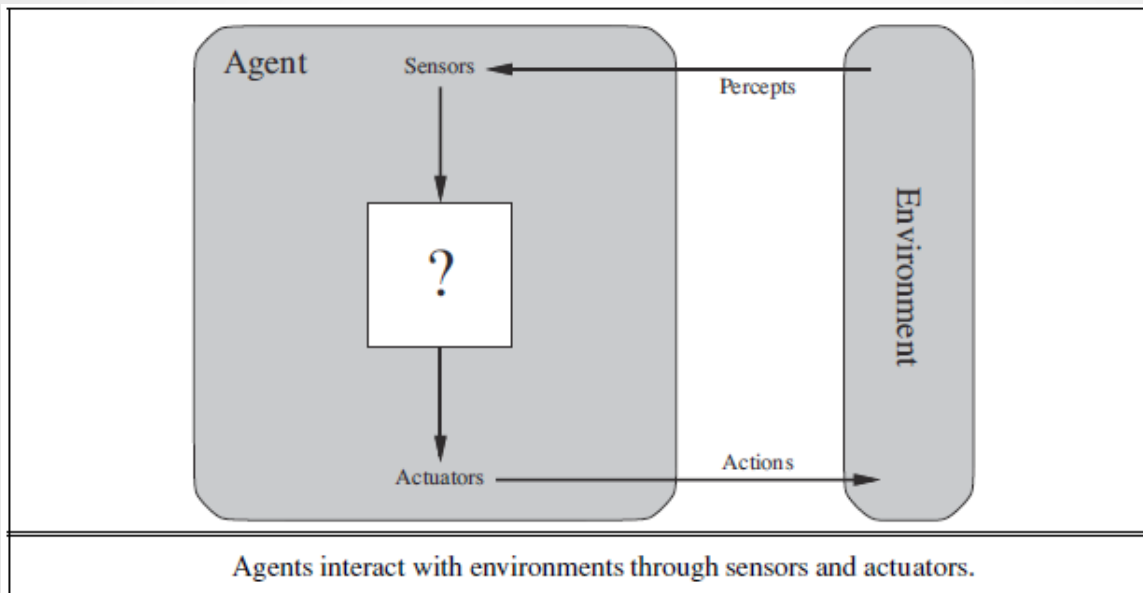
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- A **human agent** has eyes, ears, skin, etc., **for sensors**; and hands, legs, vocal tract, etc., **for actuators**.
- A **robotic agent** might have cameras, infrared range finders **for sensors**; and various motors **for actuators**.
- A **software agent** has keystrokes, input files, receiving network packets **for sensors**; and screens, output files, and sending network packets **for actuators**.

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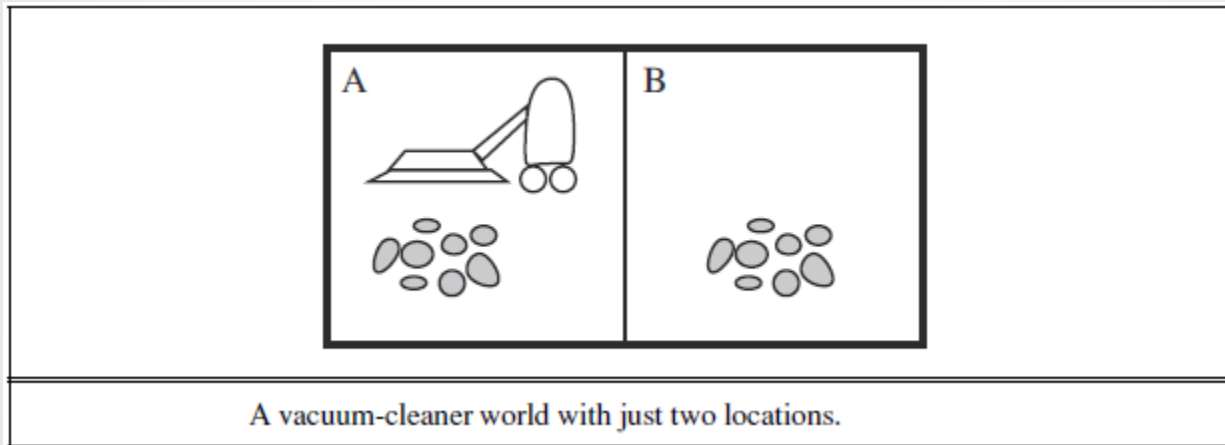
We use the term **‘percept’** to refer to the agent’s perceptual inputs at any given instant. An agent’s **‘percept sequence’** is the complete history of everything the agent has ever perceived.

Agents and Environment

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A Very Simple Example: A vacuum-cleaner



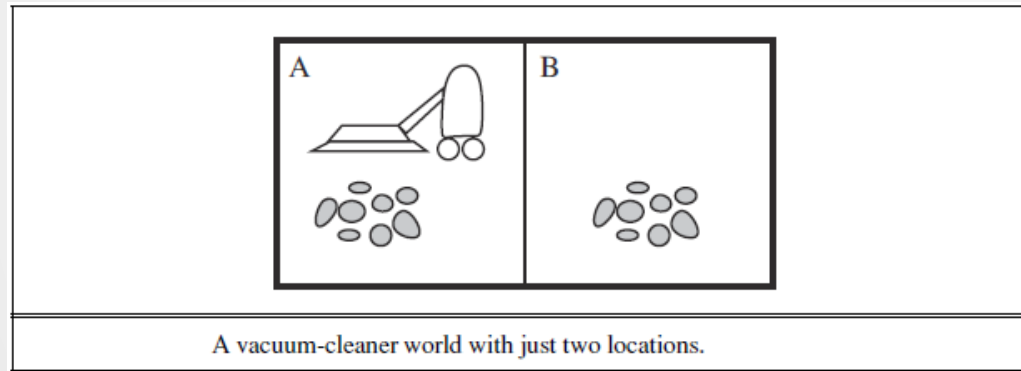
This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to *move left*, *move right*, *suck up the dirt*, or *do nothing*.

Agents and Environment

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- An agent's behavior is described by an **agent function** that maps any given percept sequence to an action.



A simple agent function: If the current square is dirty, then suck. Otherwise, move to the other square.

- For artificial agents, the agent function will be implemented by an **agent program**. The agent function is an abstract mathematical description (also can be considered as an algorithm), while the agent program is a concrete implementation, running within some physical system (ex: a computer program).

Agents and Environment

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A **rational agent** is one that does the right thing.



What does it mean to do the right thing?

The sequence of actions causes the environment to go through a **sequence of states**. A **state** can be considered as a complete picture of the environment in an instant.

The sequence of
actions



A sequence of (environment) states



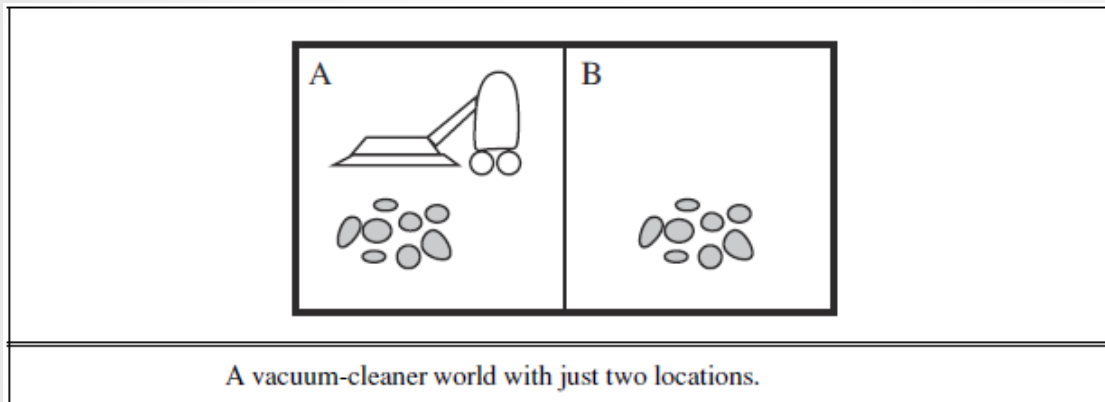
If the sequence of **environment states** is desirable, then the agent is assumed to be rational. The concept of desirability is determined by a **performance measure** that evaluates any given state of the environment.

Agents and Environment

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The performance measure



There is more than one performance measure for an agent and its environment. For the vacuum cleaner example, two of the performance measures are defined below:

1. The amount of dirt sucked up in a specific time interval.
2. One point could be awarded for each clean square at each time step.

Which performance measure is better? Why?

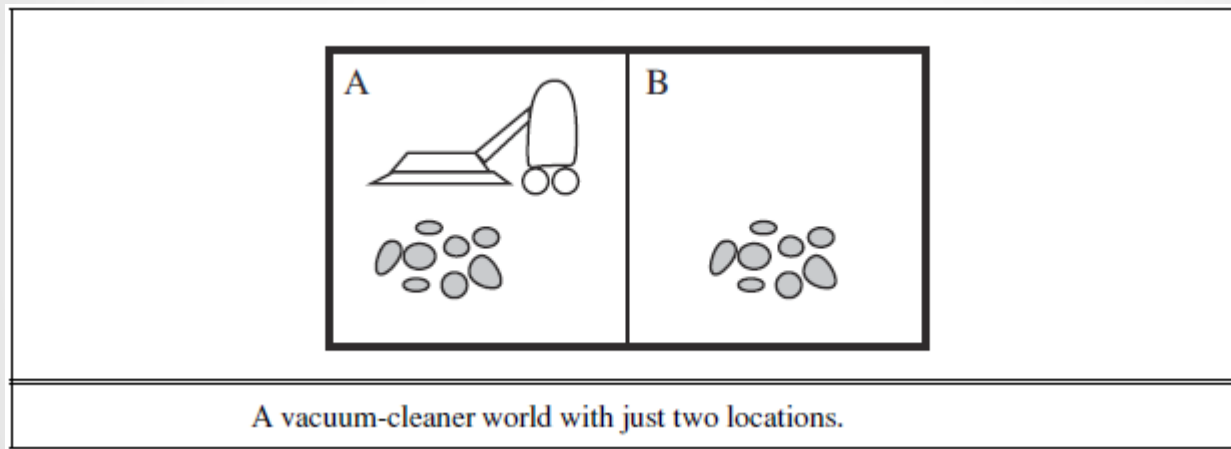
As a general rule, it is better to define performance measures according to the environment, rather than according to the behaviors of the agent.

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The performance measure



Recommend a better performance measure for a vacuum cleaner example!

Consider an environment containing noise, energy amount (the required for the agent), etc.

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Rationality

Rational behavior of an agent at any given time depends on four things:

- The performance measure (that defines the criterion of success.)
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

This leads to a **definition of a rational agent**:

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and the a priori knowledge the agent has.

PEAS

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PEAS

Before designing an agent, we should first determine the factors that are called **PEAS**.

PEAS:

- **P**erformance Measure
- **E**nvironment
- **A**ctuators
- **S**ensors

PEAS

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Example: PEAS for Autonomous Taxi Driver

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard
PEAS description of the task environment for an automated taxi.				

PEAS

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Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry
Examples of agent types and their PEAS descriptions.				

The Nature of Environment

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For the design of an agent, **the nature (properties) of the environment** should be considered.

- **Fully observable vs. partially observable:** If an agent's sensors perceive the complete state of the environment at each point in time, it is fully **observable**. Otherwise, it is **partially observable**. Rarely, the environment may be **unobservable**, but the agent's goals may still be achievable.
- **Single-agent vs. multi-agent:** If there are other agents that try to maximize their performance measure, the environment contains **multi-agent**. Here, an agent's performance measure depends on another agents' behavior. Other agents can be competitive, cooperative or partially competitive.

The Nature of Environment

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- **Deterministic vs. stochastic:** If the next state of the environment is completely determined by the current state and the action, then the environment is **deterministic**; otherwise, it is **stochastic** or **non-deterministic**. If the probability of uncertainty is known, it is called stochastic, otherwise non-deterministic.
- **Episodic vs. sequential:** In an episodic environment, the agent's experience is divided into atomic episodes. In each episode, the agent receives a percept and then performs a single action. The current episode does not depend on the actions taken in previous episodes. In sequential environments, on the other hand, the current decision could affect all future decisions.

The Nature of Environment

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- **Static vs. dynamic:** If the environment can change while an agent is deciding the process, then the environment is **dynamic** for that agent; otherwise, it is **static**. If the environment itself does not change with the passage of time but the agent's performance score changes, then we say the environment is **semi-dynamic**.
- **Discrete vs. continuous:** The **discrete/continuous** distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent.

The Nature of Environment

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The **hardest case** is *partially observable, multi-agent, stochastic, sequential, dynamic and continuous.*

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete
Examples of task environments and their characteristics.						

The Structure of Agent Types

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The Structure of Agents

The job of AI is to design an **agent program** that implements the **agent function**. We assume this program will run on some computing device with physical sensors and actuators. We call this the **architecture**.

$$\textit{agent} = \textit{architecture} + \textit{agent program}$$

How can an agent program be developed?

By using **the table-driven approach**?

The Structure of Agent Types

INTRODUCTION TO ARTIFICIAL INTELLIGENT



A table-driven for the vacuum-cleaner

Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
<i>[A, Clean], [A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
Partial tabulation of a simple agent function for the vacuum-cleaner world	

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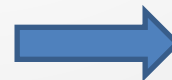
A Table-Driven Approach

Let $\wp$ be the set of possible percepts and let T be the lifetime of the agent (the total number of percepts it will receive). The lookup table will contain

$$\sum_{t=1}^T |\wp|^t$$

entries.

Consider the automated taxi: the visual input from a single camera comes in at the rate of roughly 27 megabytes per second (30 frames per second, 640×480 pixels with 24 bits of color information)



$> 10^{250,000,000,000}$
for one hour's driving

percepts

The Structure of Agent Types

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A Table-Driven Approach

The estimated number of atoms in the observable universe is less than 10^{80} .

means that



No physical agent in this universe will have the space to store the table.

A table-driven approach fails!

The key challenge for AI is to find out how to write programs that produce rational behavior from a small program rather than from a vast table.

For Example:

The large tables of square roots are used by engineers and school children up to 1970s.

But now



We can use Newton's Method, which can be applied with a five-line code, to compute square roots.

The Structure of Agent Types

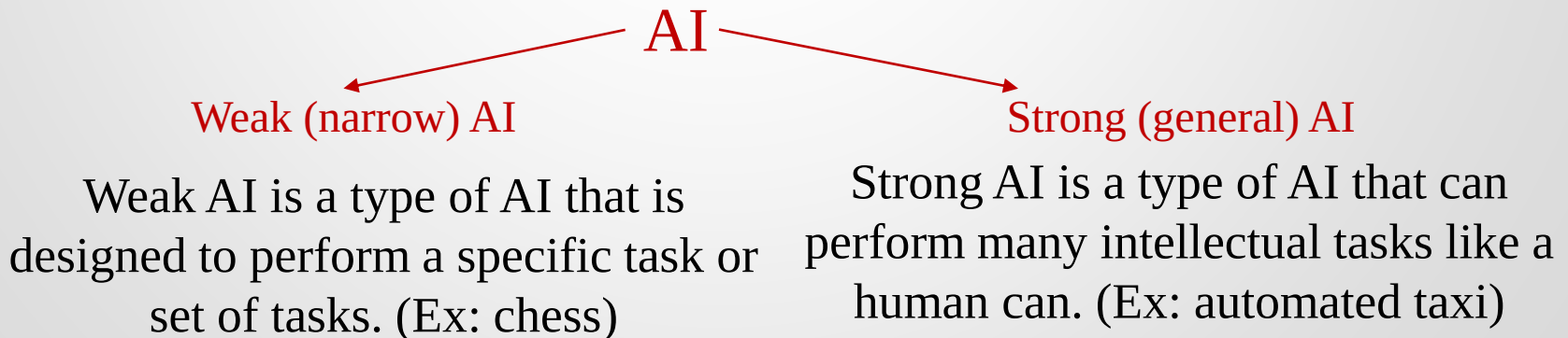
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A small program for the vacuum cleaner

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Vacuum cleaner is an example of weak AI.



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Four Basic Types of Agent Programs

There are four basic types of agent programs for almost all intelligent systems:

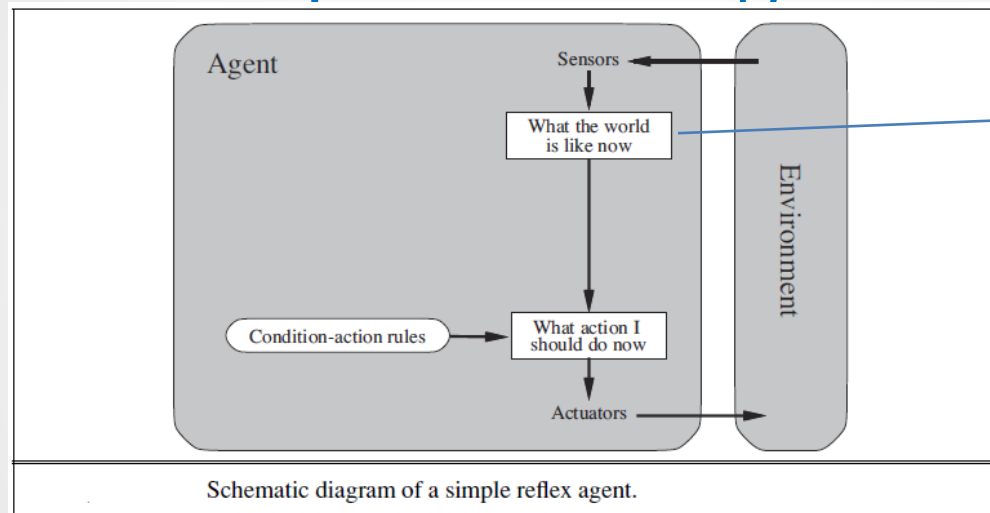
- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents

The Structure of Agent Types

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Simple Reflex Agents



the internal state,
which reflect the
environment state.

These agents select actions on the basis of the current percept, ignoring the rest of the percept history (it does not remember its past!).

Ex: vacuum cleaner

condition-action rule: if-then rules

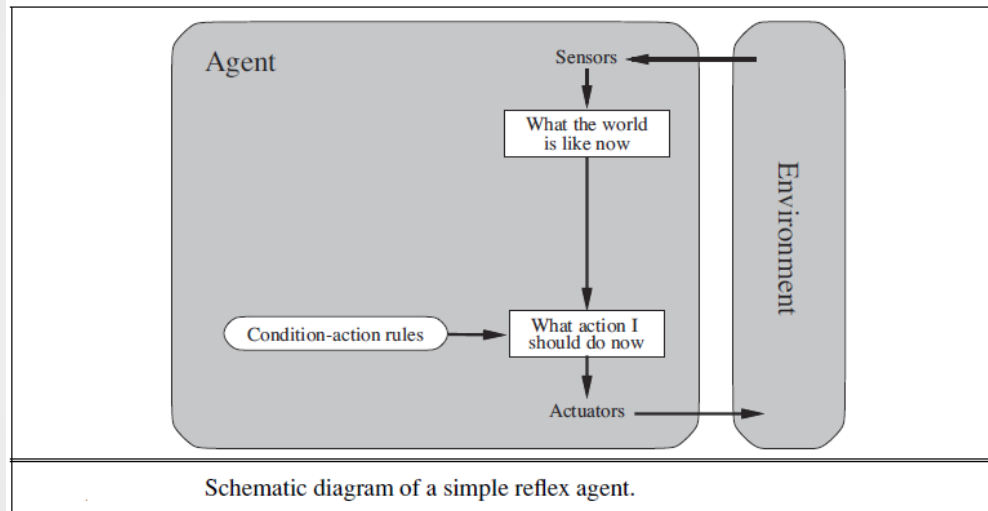
Ex: **if** *car-in-front-is-braking* **then** *initiate-braking*

The Structure of Agent Types

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Simple Reflex Agents



Especially, when the environment is partially unobserved, simple reflex agents can be in trouble. They also do not remember their past so they cannot estimate the unobserved parts.

Ex: Consider a vacuum cleaner that cannot perceive its location. Thus,

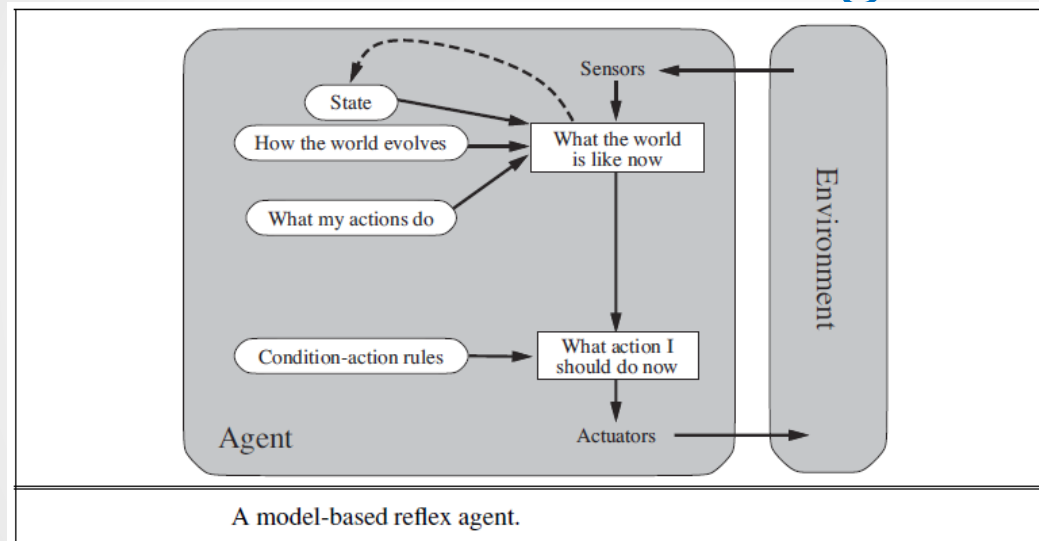
if *square-is-clean* **then** (*move left or right ??*)

The Structure of Agent Types

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Model-Based Reflex Agents



The knowledge about “**how the world works (independently of agents itself)**” or “**how the agent’s own actions affect the world**” is called a **model** of the world. An agent that uses such a model is called a **model-based agent**. It remembers its past, since it stores the set of internal states corresponding to the environment states that it encounters.

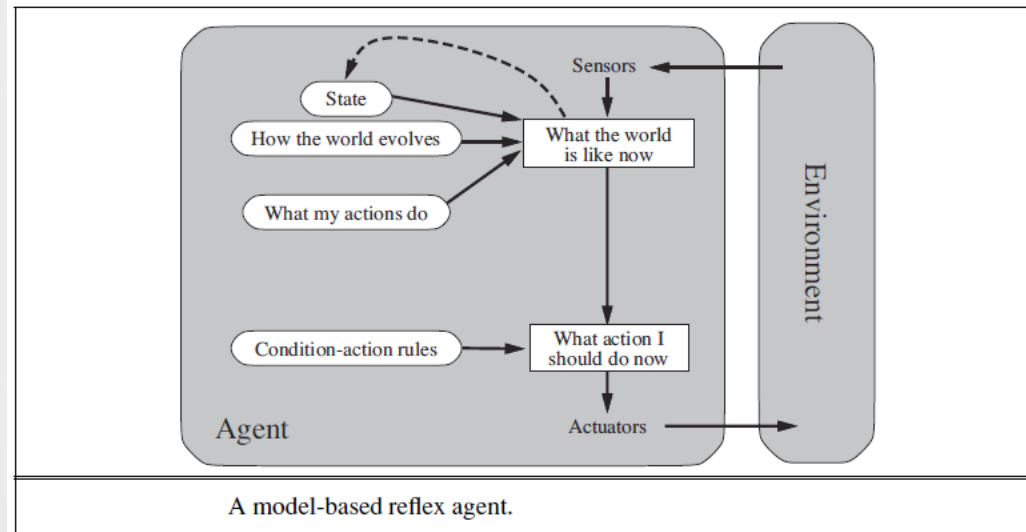
Ex: an overtaking car generally will be closer behind than it was a moment ago (it knows how the world works), or when the agent turns the steering wheel clockwise, the car turns to the right (it knows how the agent’s own actions affect the world)

The Structure of Agent Types

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Model-Based Reflex Agents



Ex: Consider the vacuum cleaner that cannot perceive its location.

if *square-is-clean* **then** (*move left or right ??*)

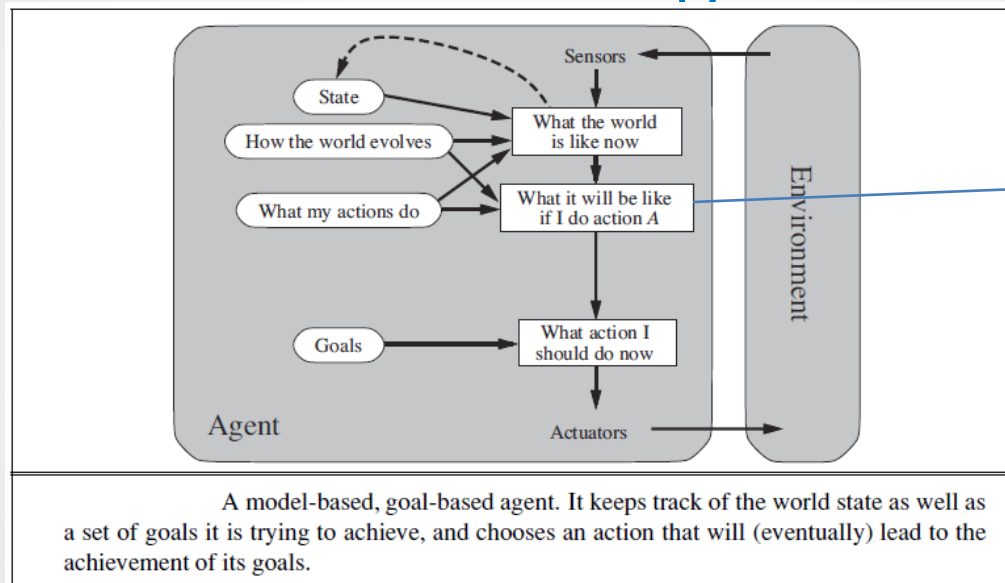
Since a model-based reflex agent remembers its past, it can choose the correct action, unlike simple reflex agents.

The Structure of Agent Types

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Goal-Based Agents



It can consider the possible future states according to each action.

As well as a current state description, the agent needs some sort of **goal** information that describes situations that are desirable. It requires **search** and **planning**, which are the subfields of AI.

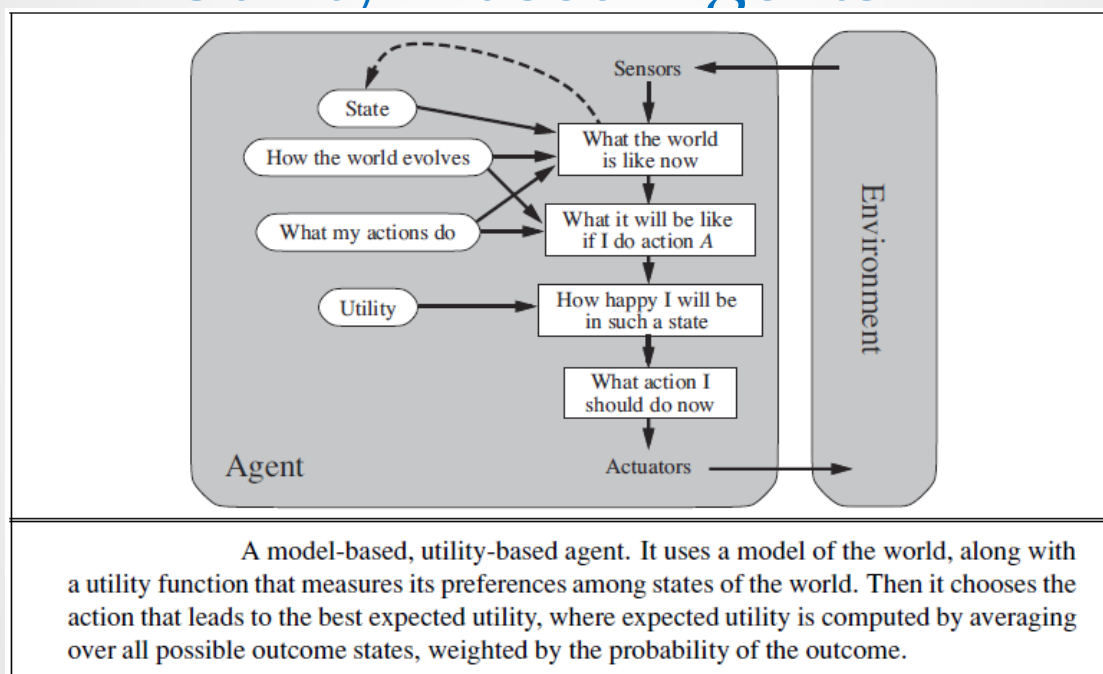
Ex: For the taxi, the goal is being at the passenger's arrival location

The Structure of Agent Types

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Utility-Based Agents



Goals alone are not enough to generate high-quality behavior in most environments. Also, whether the agent is happy sufficiently should be considered.

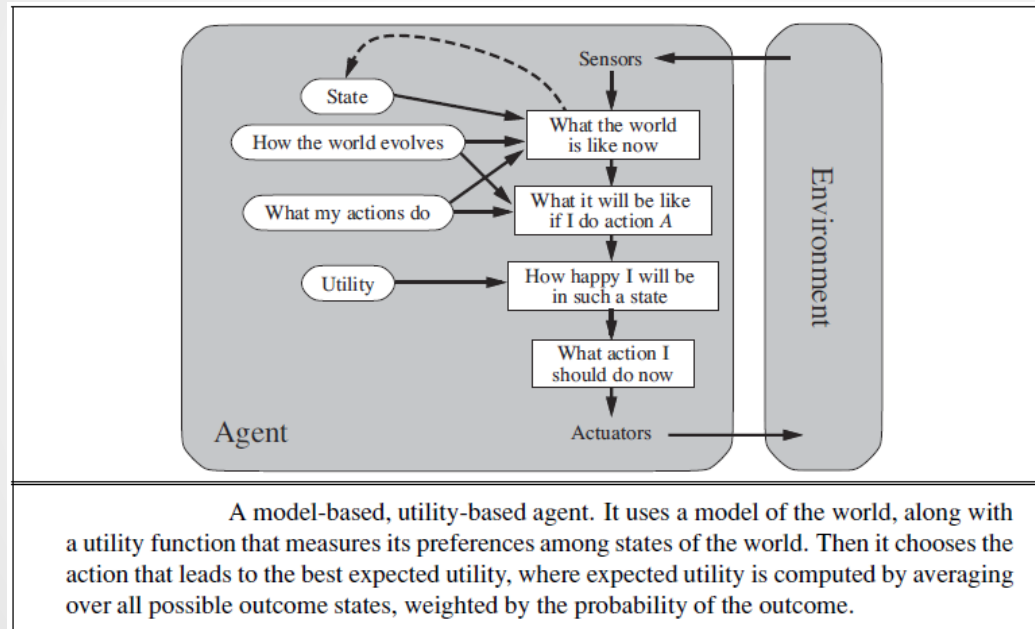
Ex: Many action sequences will get the taxi to its destination (**goal**) but some are quicker, safer, more reliable, or cheaper than others (**utility**).

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Utility-Based Agents



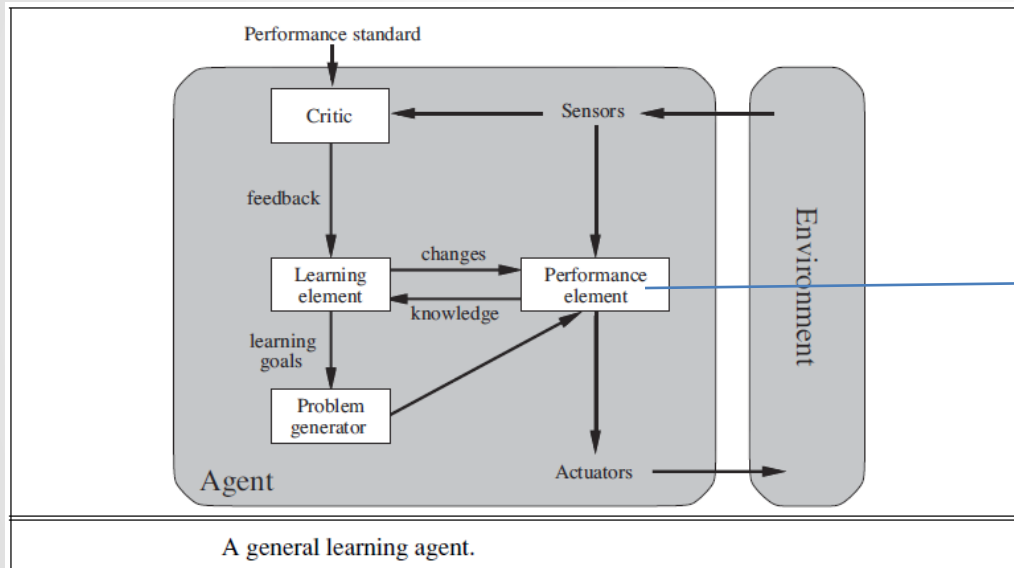
An agent's **utility function** is essentially an internalization of the **performance measure**. It means that the agent uses the performance measure during the decision process, and this measure is called the **utility function**.

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Learning Agents



One of the simple reflex agent, model-based reflex agent, goal-based agent or utility-based agent

A learning agent is composed of four components:

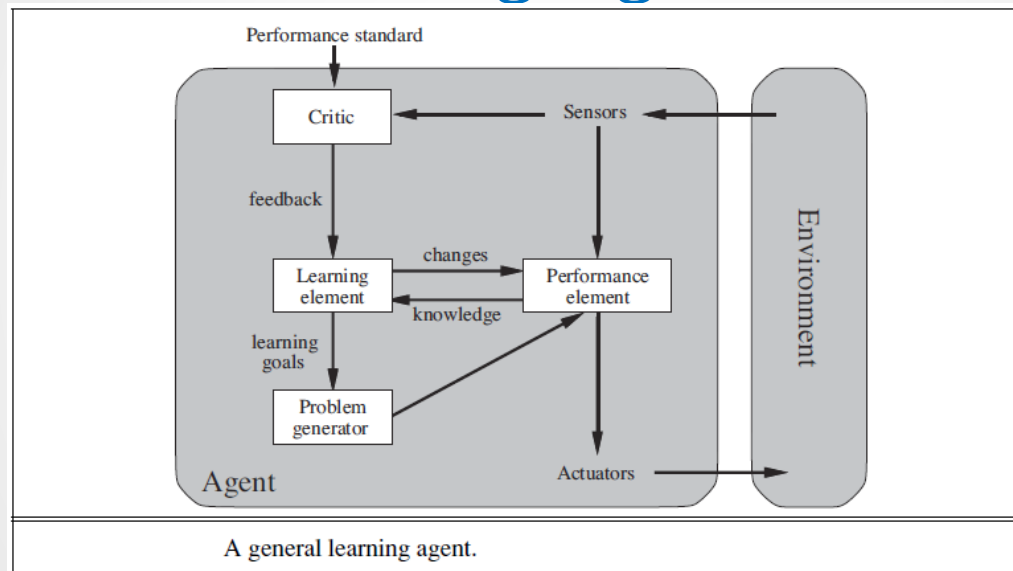
1. **Performance element** is responsible for selecting actions. It is one of the agents that we have discussed above.
2. **Learning element** is responsible for making improvements.

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Learning Agents



3. **Critic:** The learning element uses feedback from the **critic** on how the agent is doing and determines how the performance element should be modified to do better in the future.

4. **Problem generator** is responsible for suggesting actions that will lead to new and informative experiences.

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Learning Agents

For the taxi example:

Suppose that the taxi agent drives by using its **performance element**.

Critic: After the taxi makes a quick left turn across three lanes of traffic, the critic observes the shocking language used by other drivers. From this experience, the **learning element** formulate a rule saying this was a bad action, and accordingly, the performance element is modified. Thus, next time it will not make a quick left turn across three lanes of traffic.

The problem generator: The agent tries out the brakes on different road surfaces under different conditions and learns the stopping distance for different road surfaces.

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Main Reference

Stuart Russell, Peter Norvig; “Artificial Intelligence A Modern Approach”, Pearson, 3rd Edition, 2013.